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Investigation on Graphene Nanofluids and its Applications: A brief Literature Review

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S. Suneetha¹ and P. Bala Anki Reddy^{2*}¹Department of Applied Mathematics, Yogi Vemana University, Kadapa-516003 India.²Department of Mathematics, S.A.S., VIT University, Vellore-632014 India.*Corresponding Author E-mail: pbarmaths@gmail.com**ABSTRACT:**

In this review, the heat transfer capabilities of graphene nano fluids have been explored theoretically and experimentally. This review embraces the experimental results about the capabilities of graphene nano fluids along with heat transfer performance and recaps the recent growth on preparation and evaluation methods, the ways to enhance the stability of graphene nano fluids and future applications in various fields of energy. Moreover, this review paper also specifies the inconsistencies among them. Further, this critical review helps the researchers to investigate the heat transfer on graphene nanofluids embedded with conducting dust particles.

KEYWORDS: Thermal Conductivity, Nanofluids, Nano particles, Heat transfer, graphene nanoparticles.**INTRODUCTION:**

Nowadays, the industrial technology developments are mainly focussing on the optimization of energy and in reducing the processing time of a system. As a result, the ultra-high performance in cooling and heating of the thermal systems become an essential in the heat transfer field. In several engineering applications, heat transfer along a fluid medium is significant in automobiles, heat exchangers, refrigerators and power plants. Heat transfer through a fluid medium is important in several engineering applications, including heat exchangers, refrigerators, automobiles and power plants. The capacity of a fluid medium to transfer a huge quantity of heat over a small temperature difference increases the effectiveness of energy conversion and upgrades the design and performance of automobile engines, heat transfer devices and micro-electro-mechanical systems (MEMS). Heat transfer through fluid is basically convection control. However, the thermal conductivity of the fluid strongly depends on the coefficient of convective heat transfer.

Thermal conductivity can be improved by suspending solid particles in the liquid medium is an effective approach as thermal conductivity of solids is greater than that of fluids. A nano fluid is a new class of solid liquid mixtures of stable colloidal suspension of nanometre size particles i.e., multivalued carbon nanotubes which enhances the conductivity of the fluid. This nanofluid theory was introduced by Choi [1995]. Nano fluids are the most potential heat transfer fluid used as a coolant [Sharma et al. (2016), Chandra Prakash (2015)]. Nanofluid is not only the mixture of liquid-solid alloys. It is also having some special characteristics like stable suspension, durability and collection of particles. Due to the fascinating thermal characteristics of nanofluids, this research has attracted a vast interest from the researchers worldwide.

Fig.1 Preparation of nanofluid and its structure

Nanofluids are the suspensions of the nano meter sized metallic or non-metallic powders into base fluids. Fig.1 shows the pictorial presentation of the preparation of a nanofluid and cross section of nanofluid structure consisting of nano particles.

Convectional base fluids contain mm or micro meter sized particles. Due to this reason, the thermal conductivity of the convectional base fluids has inherently poor while compared with the solids. The suspension of nano powders into convectional base fluids helps to enhance the heat transfer rate. Table 1 depicts the materials for base fluids and nano particles.

Table.1 Materials for base fluids and nano particles

Base Fluids	Nanoparticles material
Water	Oxide Ceramics – CuO, Al ₂ O ₃ , ZrO ₂ , TiO ₂
Ethylene or try-ethylene glycols	Metals – Al, Cu, Zn, Ag
Polymer solutions	Nitrides – A/N, SiN
Bio fluids	Metal Carbides – SiC
Oil and other lubricants	Non-metals – Graphite, carbon nanotubes
Other common fluids	Phase change materials and Functionalized nano particles

Nanofluids can be produced in various methods. Mainly they are of two types: 1) One-step and 2) Two step methods. Preparation of nanofluid by one-step method [Lee et. al (2012), Kumar et. al (2009), Eastman et.al (2001), Zhu et. al (2004)], both the preparation of nano particle and production of affects are made out simultaneously in a collective process. This method has a few advantages like minimizing the agglomeration of nano particle. But some production methods are costly and troublesome.

The most commonly used method is the two-step method [Abbasian et al. (2012), Yu et al.(2009)], nano particles, nanotubes, nanofibers or any nanostructured materials are primarily symphonized as dry powders by physical or chemical methods. Then the dry powders are distributed in the conventional base fluids. Conversely, because of high surface activity of particles, agglomeration/aggregation of nano particles is inevitable. Consequently sedimentation of nano particles can happen in these methods, which influence the nanofluid properties negatively. This is the disadvantage in the two-step method.

From the previous literature, nano particles could be metallic, non-metallic compounds as Ag, Cu, Ni, Au, Fe, etc., ceramic compounds such as oxides(Al₂O₃, Fe₃O₄, ZnO, SiO₂, Fe₂O₃, CuO, TiO₂, CeO₂), sulfides (WS₂, MoS₂) and carbides (SiC). Nano particles can be also carbon based compounds, like carbon nanotube [Xu et al.(2012)], graphene [Gupta et.al(2011)], graphene oxide [Park et al. (2010)], graphite [Sun et al. (2013)] and also nanoscale liquid droplets. Base fluids are certain from water, ethylene glycol, mixture of water and ethylene glycol, diethylene glycol, polyethylene glycol, engine oil [Ettefaghi et al. (2013)], vegetable oil [Alves et al. (2013)], paraffin [Khedkar et al. (2013)], coconut oil [Rashin et al. (2013)], gear oil [Kole et al. (2011)], kerosene [Yu et al. (2010)], pump oil [Hashemi et al. (2012)], etc.

Figure.2 Schematic of Nanofluids as a three phase suspension system

This dispersion systems can be considered with three phase system as figure.2 shows: solid phase, solid/liquid interface and the liquid phase.

Thermal conductivity denotes the ability of heat transfer and is most important in thermal applications. Thermal conductivity ratios, defined as thermal conductivity of the nanofluid (k_{nf}) divided by the thermal conductivity of the base fluid (k_f). By using the carbon based nano structures in nanofluids, the heat transfer coefficient may be increased [Moghaddam et al. (2013)]. Choi et al. [2001] analyzed the heat transfer characteristics of the nanofluid by considering the different types of nano fluids and found a large improvement in effective thermal conductivity of the fluids containing metallic nano particles. Fig.3 displays the thermal conductivity of some typical materials and base fluids.

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